

Dr Umaru Adamu, PhD

Cybersecurity Specialist – Ransomware & Critical Infrastructure Protection

AI-Driven Threat Detection | OT-Aware Security | Risk & Compliance

📍 Bradford, UK |

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GitHub: <https://github.com/Umaruadamu>

PROFESSIONAL SUMMARY

Cybersecurity specialist with a PhD in Computer Science and 10+ years' experience in regulated, safety-critical infrastructure environments. Expertise in ransomware detection, malware analysis, and AI-driven threat modelling, with a strong foundation in engineering systems and operational resilience.

Experienced in network security, vulnerability assessment, and incident analysis, with the ability to translate cyber threats into operational and business risks. Actively developing capabilities in Operational Technology (OT) and Industrial Control Systems (ICS) security, with a focus on protecting energy and Oil & Gas infrastructure from ransomware and advanced cyber threats.

CORE SKILLS

- Ransomware Detection & Threat Modelling
- Malware Analysis & Classification (Deep Learning)
- Critical Infrastructure (CNI) Security
- Cyber Risk Management & ISO 27001
- OT / ICS Security Awareness
- Network Security & Traffic Analysis (Wireshark)
- Vulnerability Assessment (Nmap, Kali Linux)

- Security Monitoring & Incident Response
- Systems Engineering & Infrastructure Resilience

PROFESSIONAL EXPERIENCE

Nigeria Atomic Energy Commission

Scientific Officer I | 2012 – 2025

Worked within a nationally regulated nuclear infrastructure environment, supporting system resilience, compliance, and operational risk management.

Key Responsibilities:

- Conducted infrastructure systems analysis and modelling
- Supported risk assessment, compliance, and audit processes
- Produced technical reports on system performance and risk
- Delivered training and coordinated staff development
- Collaborated across engineering and IT functions

Relevance to Ransomware & Energy Security:

- Experience in protecting safety-critical systems
- Understanding of operational disruption risks (key in ransomware scenarios)
- Exposure to regulated infrastructure security environments

University of Bradford

Cybersecurity Laboratory & Teaching Support | 2018 – Present

Delivered practical cybersecurity training aligned with real-world attack scenarios.

Key Activities:

- Conducted labs in:
 - Network traffic analysis (Wireshark)
 - Vulnerability scanning (Nmap)
 - Penetration testing (Kali Linux, Metasploit)
- Simulated SOC environments and threat detection workflows
- Guided students on risk assessment and ISO 27001 compliance

Ransomware-Relevant Impact:

- Practical exposure to attack detection and analysis workflows
- Simulation of threat scenarios including malware behaviour

EDUCATION

PhD – Computer Science (Cybersecurity)

University of Bradford | 2023

Deep Learning-Based Framework for Malware Detection and Classification

MSc – Advanced Control & Systems Engineering

University of Sheffield | 2016

BEng – Electrical & Computer Engineering

Federal University of Technology Minna | 2008

RANSOMWARE & CYBERSECURITY PROJECTS

Ransomware Detection Model

- Developed supervised machine learning model achieving **88.2% accuracy**
- Focused on early-stage ransomware behaviour prediction
- Applied feature engineering techniques for threat classification

AI Malware Detection Framework (PhD Research)

- Built deep learning model achieving **99.3% accuracy**
- Applied to large-scale malware datasets
- Designed scalable detection framework using ensemble techniques

Cybersecurity Lab (Attack Simulation)

- Simulated malware and network-based attacks
- Analysed suspicious traffic using Wireshark
- Conducted vulnerability assessments and exploitation testing

(In Progress) ICS/SCADA Ransomware Risk Lab

- Developing PLC simulation using OpenPLC
- Exploring ransomware impact on industrial systems
- Investigating segmentation and mitigation strategies

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Google Cybersecurity Professional Certificate
- CompTIA Security+ (In Progress)

Leadership & Development

- Leading with Vision – Strategic Leadership
- Improving Communication Using Lean Thinking
- Generative AI Foundations

TECHNICAL SKILLS

Security & Analysis

Wireshark | Nmap | Vulnerability Assessment | Threat Analysis

Programming & AI

Python | TensorFlow | PyTorch | SQL

Systems & Platforms

Azure | Docker | Virtualisation

PROFESSIONAL MEMBERSHIPS

- IEEE – Graduate Member
- Nigerian Society of Engineers (NSE)
- COREN Registered Engineer
- Nigerian Institute of Management

KEY DIFFERENTIATORS

- ✓ Specialisation in ransomware detection and malware classification
- ✓ Experience in safety-critical infrastructure (nuclear sector)
- ✓ Strong AI + cybersecurity + engineering combination
- ✓ Ability to link cyber threats to operational disruption
- ✓ Growing focus on OT/ICS ransomware risks in energy systems

REFEREES

Available upon request